

ABSTRACT

The central objective of this project is to implement an induction generator and conduct simulations using MATLAB R2023b Simulink. Initially, the project focuses on developing and integrating an induction generator within an arbitrary frame. This model facilitates the examination of the dynamic response of the squirrel cage induction generator when directly connected to the grid. Variations in initial speed will be explored, observing stator currents, electromagnetic torque, and rotor speed. Additionally, a comparison will be made between the Simscape induction generator and the Simulink SCIG model developed within this project.

I. INTRODUCTION

The rising global demand for energy underlines the necessity of transitioning away from fossil fuel dependence towards renewable alternatives. Within the domain of renewable energy, wind power assumes significant importance. Wind energy conversion systems rely on various generator types, including induction generators such as the squirrel cage induction generator (SCIG) and the double fed induction generator (DFIG), as well as synchronous generators like the permanent magnet synchronous generator (PMSG) and the wound rotor synchronous generator (WRSG). This study focuses exclusively on investigating the dynamic response characteristics of the squirrel cage induction generator, employing a model-based approach to explain its operational dynamics.

The Squirrel-Cage Induction Generator (SCIG) occupies a prominent position within contemporary electrical power systems, valued for its inherent simplicity, flexibility, and adaptability. Operating on the fundamental principles of electromagnetic induction, SCIGs find extensive application in renewable energy contexts, notably within wind power generation. Distinguished by their self-starting capability and ability to synchronize with the grid without necessitating elaborate control systems, SCIGs offer distinct advantages over synchronous generators. Induction generators function on the same principles as induction motors, generating AC electricity. The Squirrel-Cage Induction Generator (SCIG) is connected to the wind turbine via a gearbox, and a back-to-back full-power converter is interfaced with the SCIG stator windings. The gearbox employs gear ratios to transform the speed and torque transmitted from the rotating power source to a different device.

II. SIMULATION DATA AND RESULTS

As part of our project, we are tasked with implementing a squirrel cage induction generator (SCIG). To achieve this, we have been provided with parameters to develop the model using MATLAB R2023b. The parameters are mentioned in Table 1[1].

TABLE 1. Parameters for SCIG

Parameter	Symbol	Value
Rated Generator Apparent Power	$S_{s,R}$	2.59 MVA
Rated Output Power	$P_{s,R}$	2.30 MW
Rated Mechanical Input Power	$P_{m,R}$	2.3339 MW
Rated Line-Line RMS Voltage	$V_{s,LL}$	690 V (rms)
Rated Stator Frequency	f_s	50 Hz
Rated Rotor Speed	$n_{m,R}$	1512 rpm
Rated Initial Speed	$n_{m,init}$	1450 rpm
Number of Pole Pairs	P	2
Rated Mechanical Input Torque	$T_{m,R}$	14.74 kN.m
Stator Winding Resistance	R_s	1.102 mΩ
Rotor Winding Resistance	R_r	1.497 mΩ
Stator Winding Leakage Inductance	L_{ls}	0.06492 mH
Rotor Winding Leakage Inductance	L_{lr}	0.06492 mH
Magnetizing Inductance	L_m	2.13461 mH
Moment of Inertia	J	1200 kg-m ²
Grid Line-Line RMS Voltage	$V_{g,LL}$	690 V (rms)
Grid Frequency	f_g	50 Hz

As per these parameters, I designed the SCIG model in MATLAB R2023b [1]. The model is shown in Fig. 1.

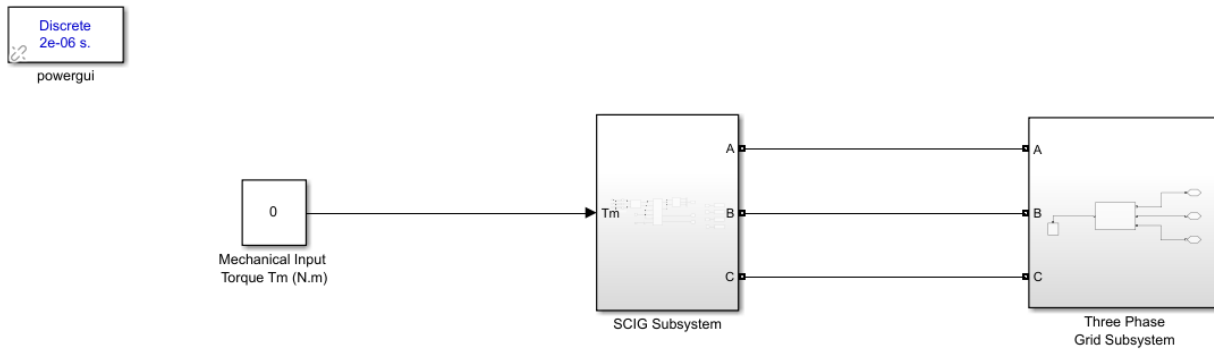


Fig. 1 Simulink Model of Squirrel Cage Induction Generator (Task-A)

In this model, I have designed two subsystems, such as the SCIG subsystem and the Three Phase Grid subsystem, as per project instructions [1].

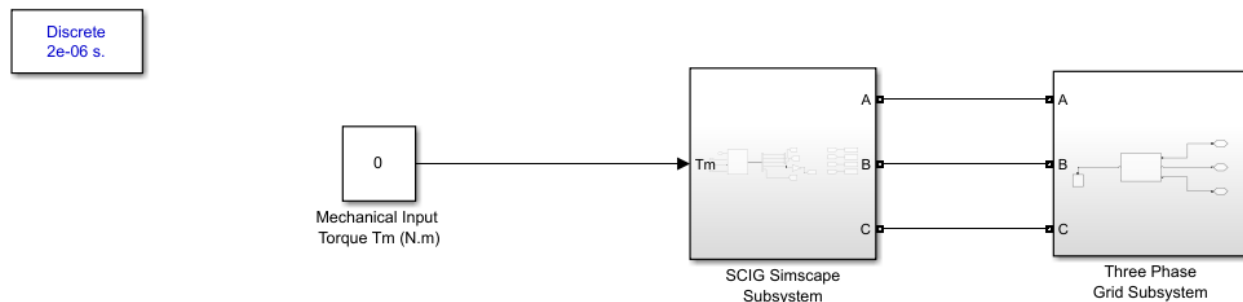


Fig. 2 Simscape Model of Squirrel Cage Induction Generator (Task-B)

In Fig. 2, I have implemented the model using the Simscape SCIG and observed the various parameters as per project instructions [1].

A. Simulation Results

Task-A: Simulink based SCIG Model

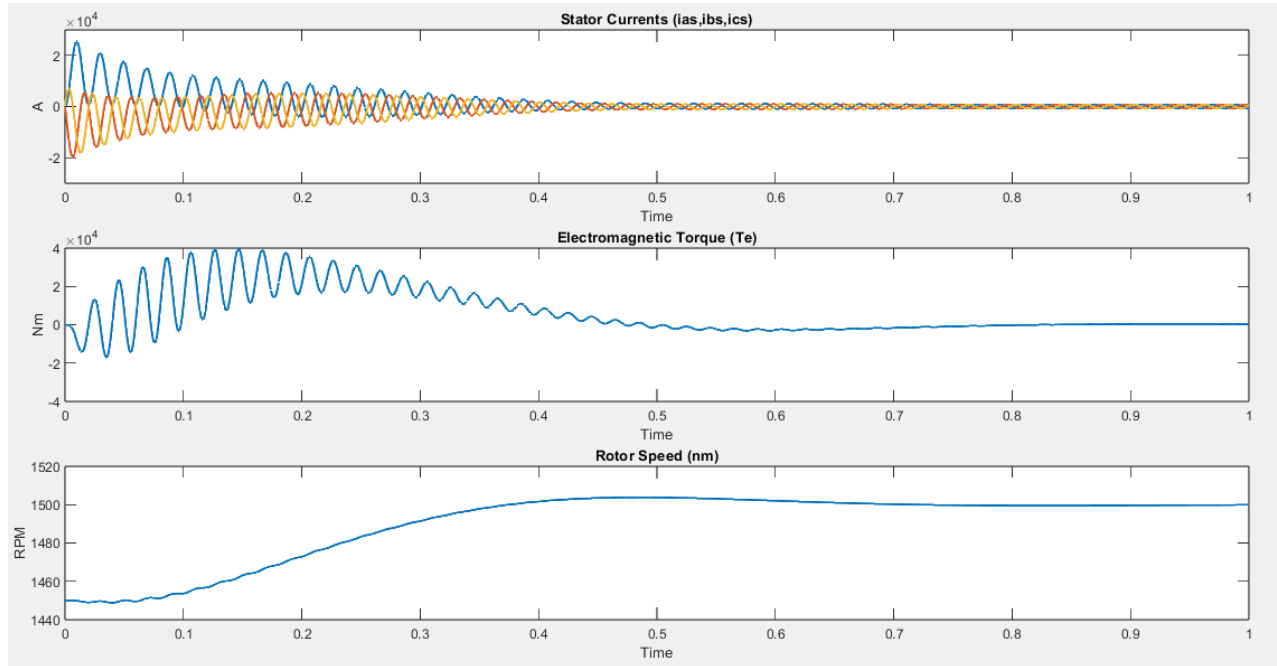


Fig. 3 Results for $n_m = 1450$ rpm

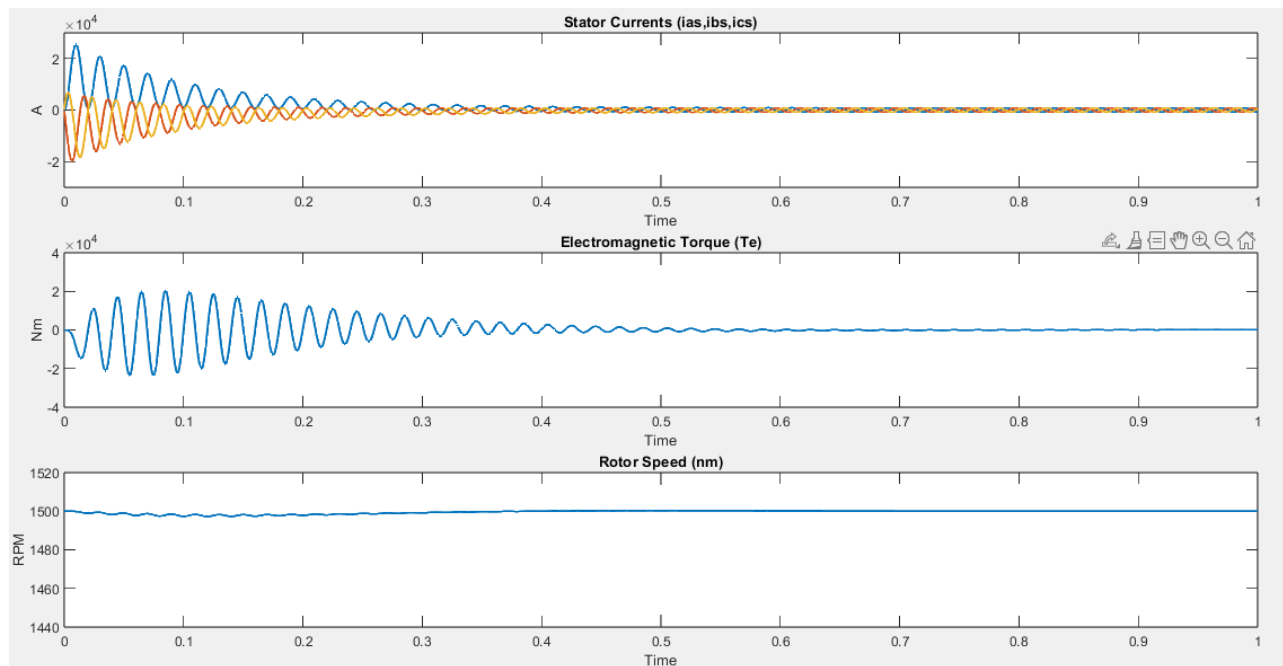


Fig. 4 Results for $n_m = 1500$ rpm

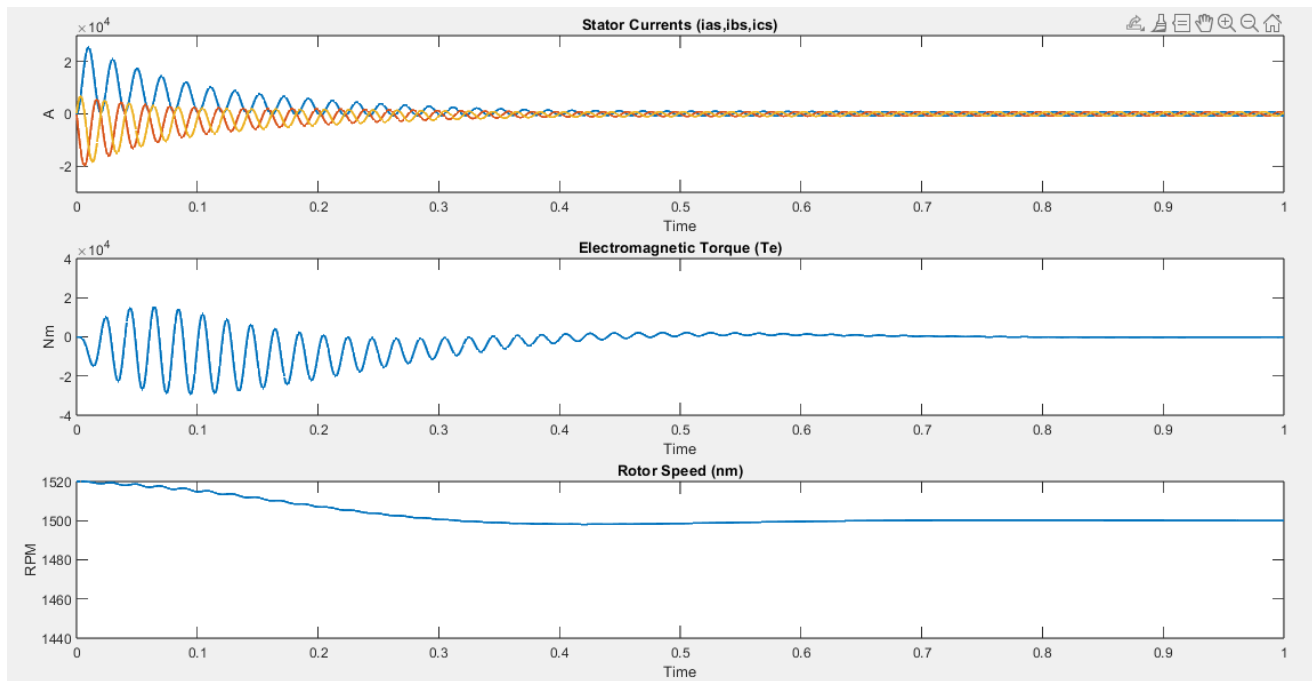


Fig. 5 Results for $n_m = 1520$ rpm

Task-B: Simscape based SCIG Model

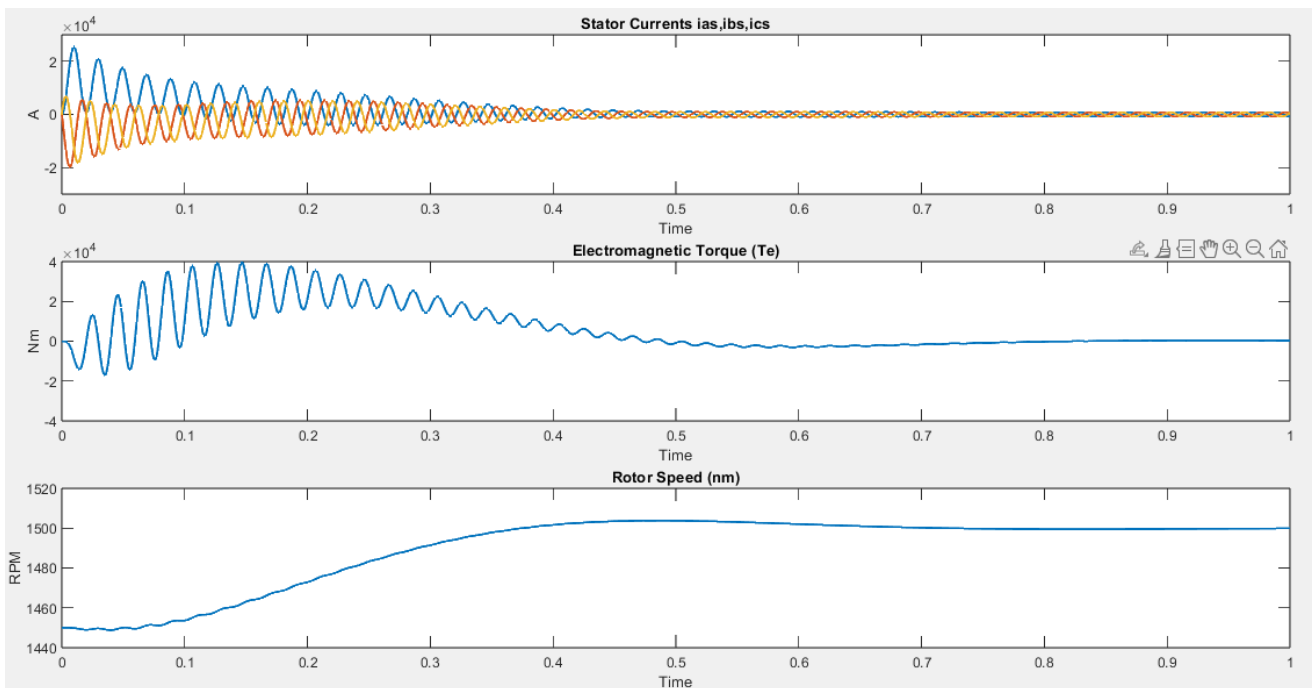


Fig. 6 Results for $n_m = 1450$ rpm

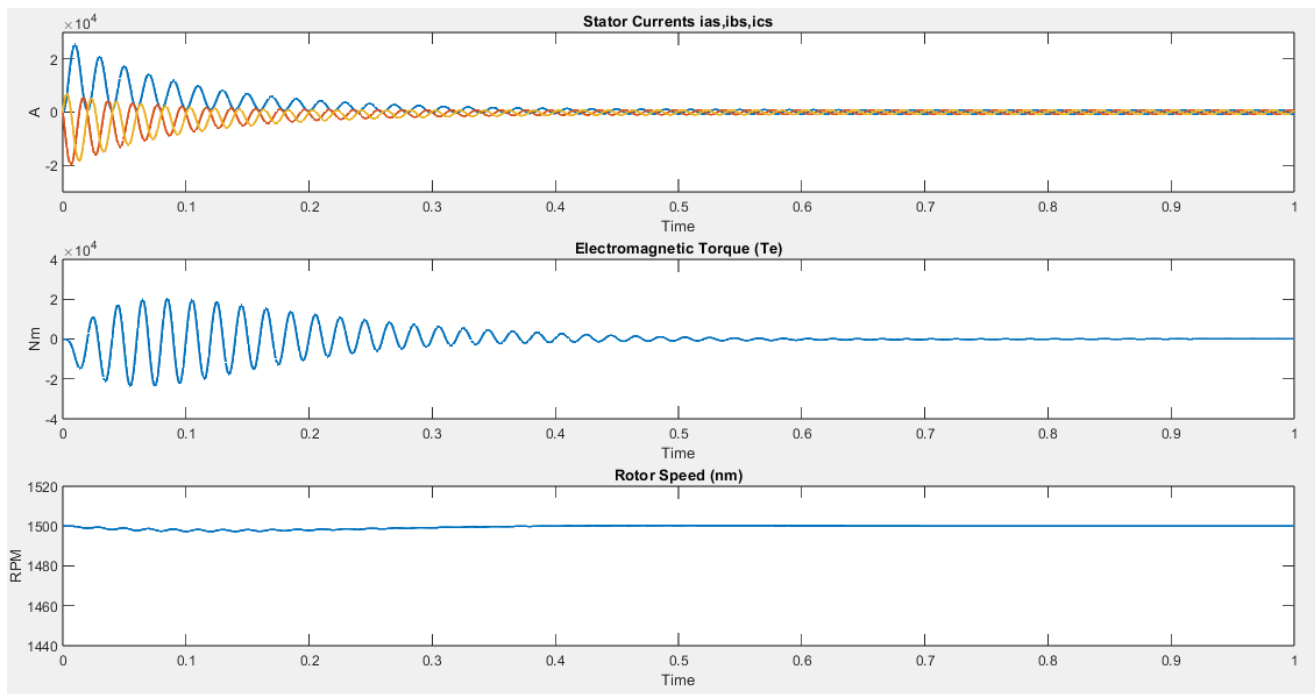


Fig. 7 Results for $n_m = 1500$ rpm

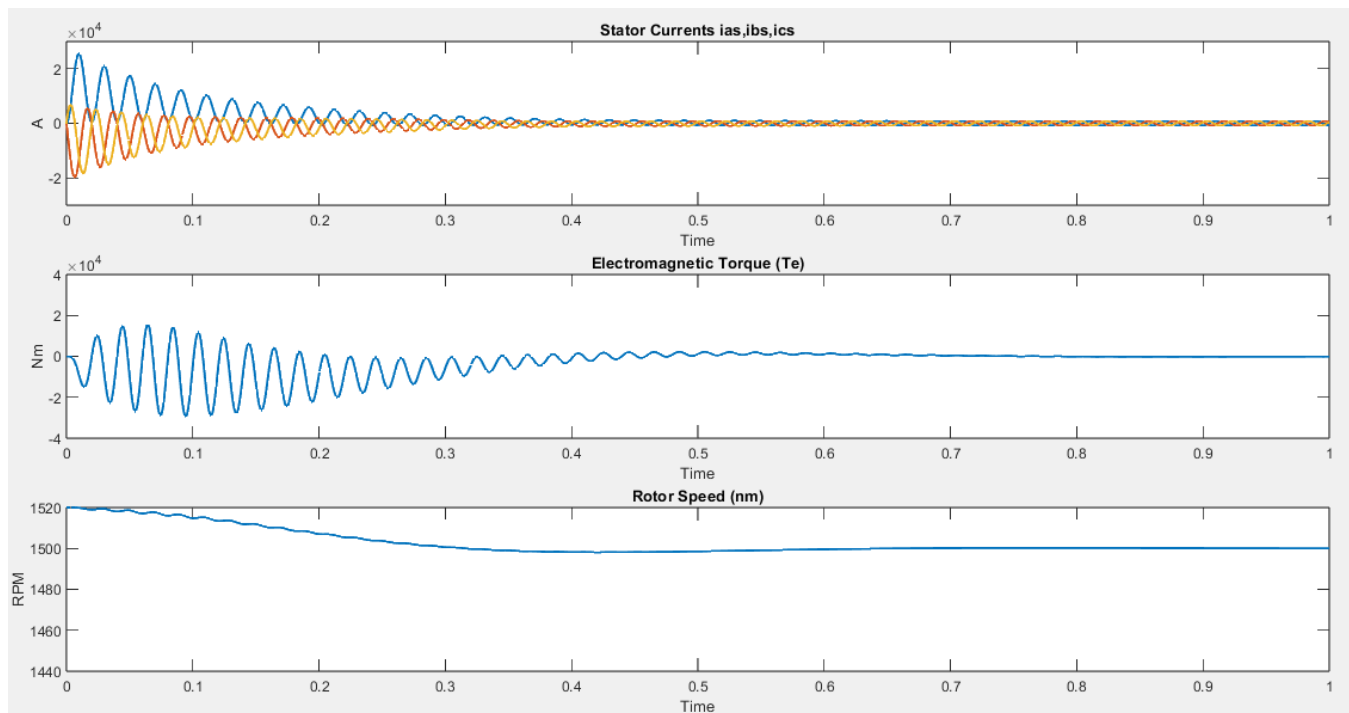


Fig. 8 Results for $n_m = 1520$ rpm

Task-3: Comparison between Simulink SCIG Model and Simscape SCIG Model

TABLE 2. Simulink SCIG Parameters

	Measured Values	$n_m = 1450 \text{ rpm}$	$n_m = 1500 \text{ rpm}$	$n_m = 1520 \text{ rpm}$
1	Maximum positive peak value among the three-phase stator currents (A)	2.544e+04	2.548e+04	2.55e+04
2	Maximum negative peak value among the three-phase stator currents (A)	-1.974e+04	-1.976e+04	-1.977e+04
3	Maximum positive peak value of the electromagnetic torque (N.m)	3.958e+04	2.035e+04	1.548e+04
4	Maximum negative peak value of the electromagnetic torque (N.m)	-1.703e+04	-2.357e+04	-2.396e+04
5	Settling maximum value of the rotor speed (rpm)	1.5038e+03	1.5003e+03	1.5200e+03
6	Settling minimum value of the rotor speed (rpm)	1.4488e+03	1.4973e+03	1.4981e+03
7	Peak time of the rotor speed (s)	0.4897	0.4906	0
8	Rotor speed overshoot (%)	0.2610	0.0187	1.3332

TABLE 3. Simscape SCIG Parameters

	Measured Values	$n_m = 1450 \text{ rpm}$	$n_m = 1500 \text{ rpm}$	$n_m = 1520 \text{ rpm}$
1	Maximum positive peak value among the three-phase stator currents (A)	2.544e+04	2.548e+04	2.55e+04
2	Maximum negative peak value among the three-phase stator currents (A)	-1.974e+04	-1.976e+04	-1.976e+04
3	Maximum positive peak value of the electromagnetic torque (N.m)	3.958e+04	2.035e+04	1.548e+04
4	Maximum negative peak value of the electromagnetic torque (N.m)	-1.703e+04	-2.357e+04	-2.396e+04
5	Settling maximum value of the rotor speed (rpm)	1.5038e+03	1.5003e+03	1.5200e+03
6	Settling minimum value of the rotor speed (rpm)	1.4488e+03	1.4973e+03	1.4981e+03
7	Peak time of the rotor speed (s)	0.4897	0.4906	0
8	Rotor speed overshoot (%)	0.2610	0.0187	1.3332

** Upon comparing the output results of Simulink and Simscape, it was observed that the two outputs are very identical, with only minor difference in the values of the negative peak three-phase stator currents at $n_m = 1520 \text{ rpm}$.

** I had observed these parameters in MATLAB using data statistics methods available in view option in figure. I had observed parameters from 5 to 8 using stepinfo function of MATLAB which I had mentioned in plot file of model.

B. Calculations

$$slip = \frac{\eta_{syn} - \eta_{init}}{\eta_{syn}}$$

Where $\eta_{init} = 1450 \text{ rpm}, 1500 \text{ rpm}$ and 1520 rpm

$$\eta_{syn} = \frac{120 \times f_s}{2P}$$

Where $f_s = 50\text{Hz}$ and $P = 2$

$$\eta_{syn} = 1500 \text{ rpm}$$

So, $slip = 0.0333, 0$ and -0.0133 respectively

Equation used during implementation of Simulink model of SCIG[2]:

Flux Linkage Equations

$$\lambda_{ds} = (v_{ds} - R_s i_{ds} + \omega \lambda_{qs}) / S$$

$$\lambda_{qs} = (v_{qs} - R_s i_{qs} - \omega \lambda_{ds}) / S$$

$$\lambda_{dr} = (v_{dr} - R_r i_{dr} + (\omega - \omega_r) \lambda_{qr}) / S$$

$$\lambda_{qr} = (v_{qr} - R_r i_{qr} - (\omega - \omega_r) \lambda_{dr}) / S$$

Where S is an integrator

Current Equations

$$\begin{bmatrix} i_{ds} \\ i_{qs} \\ i_{dr} \\ i_{qr} \end{bmatrix} = \frac{1}{D_1} \begin{bmatrix} L_r & 0 & -L_m & 0 \\ 0 & L_r & 0 & -L_m \\ -L_m & 0 & L_s & 0 \\ 0 & -L_m & 0 & L_s \end{bmatrix} \begin{bmatrix} \lambda_{ds} \\ \lambda_{qs} \\ \lambda_{dr} \\ \lambda_{qr} \end{bmatrix}$$

$$D_1 = L_s L_r - L_m^2$$

Stator and Rotor Self Inductance

$$L_s = L_{ls} + L_m$$

$$L_r = L_{lr} + L_m$$

Motion Equations

$$\omega_r = \frac{P}{JS} (T_e - T_m)$$

$$T_e = \frac{3P}{2} (i_{qs} \lambda_{ds} - i_{ds} \lambda_{qs})$$

C. Explanation

Initially, when the $n_m = 1450$ rpm, the generator will be in startup mode, due to which we will see a high rush in each stator current (i_{as} , i_{bs} , and i_{cs}) in the generator, but the sum of these stator currents will be zero due to the three-phase balanced system. As stator currents will help in magnetising the generator core and thus produce high electromagnetic torque (T_e). Also, synchronous speed is greater than initial speed, which shows that the generator is operating in motor mode. This will help in rotating the turbine during the startup of the wind energy conversion system[3].

When we set the value of $n_m = 1500$ rpm, which is equal to the synchronous speed. This is a steady-state operation condition (slip = 0, $T_e = T_m = 0$), and now the generator can start generating power by rotating blades more in the direction of wind [3].

On simulating the model with $n_m = 1520$ rpm, which is more than the base value, performing this operation will damage the generator, creating more stress on the rotor. So, it is recommended to keep the value of n_m near the base value to have good power generation [3].

III. PROJECT QUESTIONS

Q1. Why the SCIG rotor voltages v_{dr} and v_{qr} are set to zero in the given Part-A model?

Ans) As in a squirrel cage induction generator, the stator has a three-phase winding, but the rotor will have a short-circuit configuration, which results in v_{dr} and v_{qr} to zero.

Q2. What is the ω input value if the induction machine model operates in the synchronous reference frame?

Ans) Both the stator and rotor have the same type of winding in an induction machine. As a result, when an induction machine operates in a synchronous frame, the value of $\omega = \omega_s$

Q3. What will be the consequences of high inrush current and torque oscillations during direct grid connection of SCIG?

Ans) On connecting the SCIG to the direct grid, several issues may occur, such as high inrush currents, high electromagnetic torque, and high oscillations. These issues can damage the insulation of the generator winding and have a negative impact on the grid. With that, it can lead to increased mechanical stress on the generator.

Q4. From Task 2, which initial rotor speed is recommended for grid connection and why?

Ans) Basically, a base speed of 1500 rpm is recommended as its oscillation is close to 1 per unit, which helps protect the system from inrush currents and high electromagnetic torque.

Q5. Define stator and rotor self-inductances (L_s and L_r) in terms of stator and rotor leakage inductances (L_{ls} and L_{lr}) and magnetizing inductance L_m .

Ans)

$$L_s = L_{ls} + L_m \text{ (Stator Self Inductance)}$$

$$L_r = L_{lr} + L_m \text{ (Rotor Self Inductance)}$$

IV. CONCLUSION

The report details the effective deployment of a MATLAB Simulink model to simulate a squirrel cage induction generator. It explores the impact of rotor speed on SCIG while presenting a comparison of the models developed with Simscape and Simulink. The dynamic responses of SCIG were observed for both models, which came out as identical. It also helps in understanding that if the value of rotor speed goes above the base value, it may damage the generator, and keeping it close to the base value helps in generating the optimum power from the turbine.

V. REFERENCES

- [1] Prof. Apparao Dekka, "Project Instructions."
- [2] Prof. Apparao Dekka, "Lecture Slides."
- [3] B. Wu, Y. Lang, N. Zargari, and S. Kouro, *Power Conversion and Control of Wind Energy Systems*. 2011. doi: 10.1002/9781118029008.

VI. APPENDIX

1. SCIG Subsystem

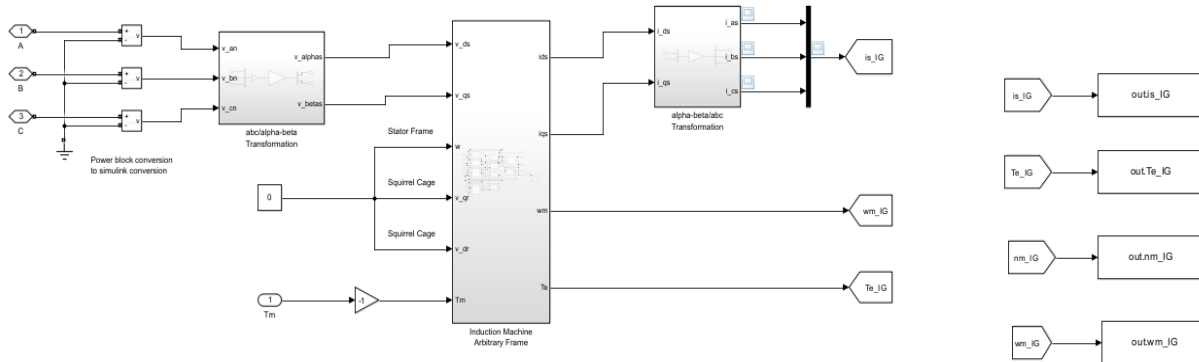


Fig. 9 SCIG Subsystem (Task-A)

2. Induction Machine Arbitrary Frame

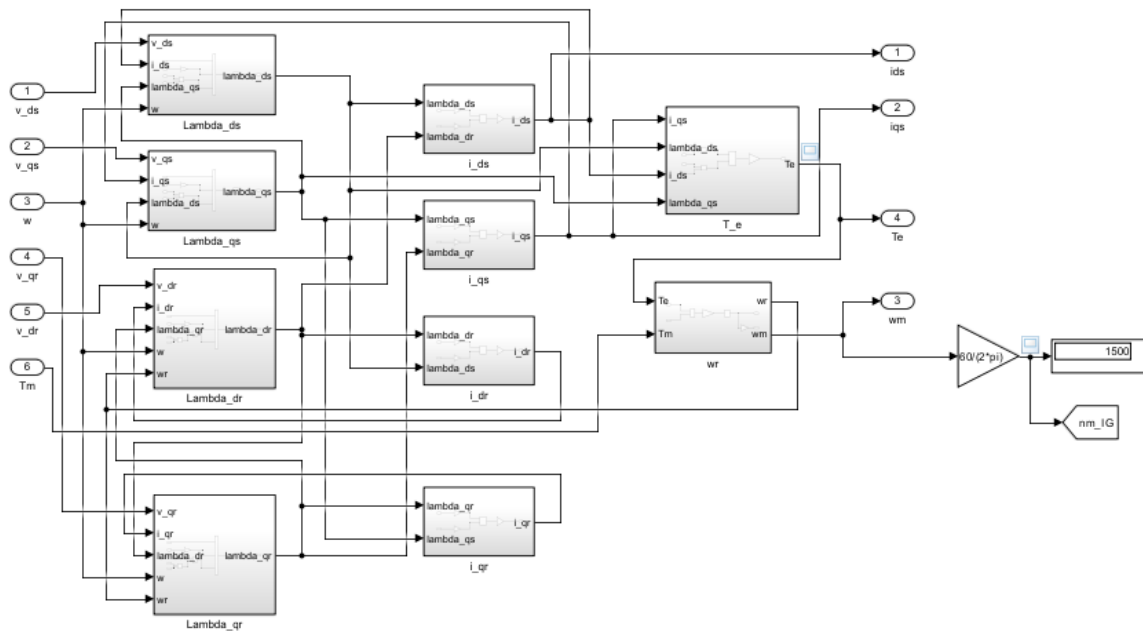


Fig. 10 Induction Machine Arbitrary Frame (Task-A)

3. SCIG Simscape Subsystem

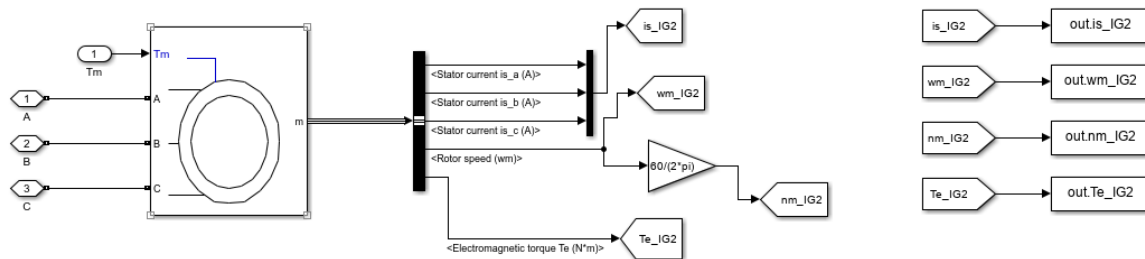


Fig. 11 SCIG Simscape Subsystem (Task-B)

4. Three Phase Grid Subsystem

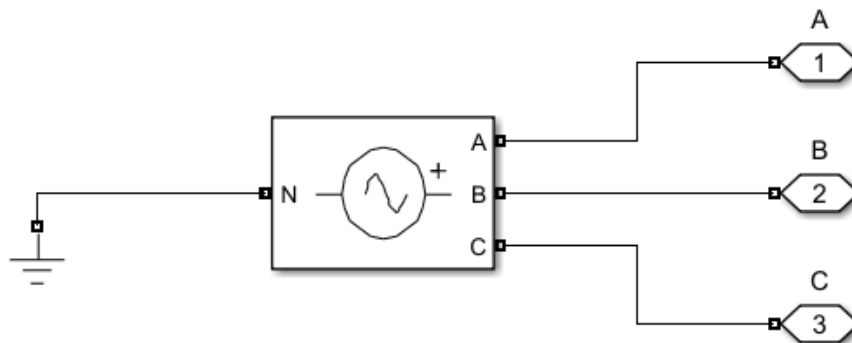


Fig. 12 Three Phase Grid Subsystem

Code:

Part 1: For Passing the constant values to Simulink Model

```

clc;
clear all;
close all;
%ABC to alpha-beta Transformation
abc_alphabeta = [1 (-1/2) (-1/2); 0 (sqrt(3)/2) (-sqrt(3)/2)];

%alpha-beta to ABC Transformation
alphabeta_abc = [1 0; (-1/2) (sqrt(3)/2); (-1/2) (-sqrt(3)/2)];

%Generator and Three-Phase Grid Parameters
  
```

```

S_sR = 2.59e06; %Rated Generator Apparent Power
P_sR = 2.30e06; %Rated Output Power
P_mR = 2.3339e06; %Rated Mechanical Input Power
V_sLL = 690; %Rated Line to Line Voltage
f_s = 50; %Rated Stator Frequency
n_mR = 1512; %Rated Rotor Speed
n_init = 1450; %Rated Initial Speed -- Change as per task in project
P = 2; %Number of Pole Pairs
T_mR = 14.74e03; %Rated Mechanical Input Torque
R_s = 1.102e-03; %Stator Winding Resistance
R_r = 1.497e-03; %Rotor Winding Resistance
L_ls = 0.06492e-03; %Stator Leakage Inductance
L_lr = 0.06492e-03; %Rotor Leakage Inductance
L_m = 2.13461e-03; %Magnetizing Inductance
J = 1200; %Moment of Inertia
V_gLL = 690; %Grid Line to Line RMS Voltage
f_g = 50; %Grid Frequency

```

```
%Calculate Determinant for current in dq axis
```

```

L_s = (L_ls + L_m);
L_r = (L_lr + L_m);
D1 = ((L_s*L_r) - ((L_m)^2));

```

```
%RPM to rad/sec
```

```
init_speed_rad = ((n_init*2*pi)/60);
```

```
%Slip Calculation
```

```

n_syn = 120*f_s/(2*P);
slip = (n_syn - n_init)/n_syn;

```

Part 2: For plotting graph for different parameters

```
clc;
```

```
%Time for Array (To Workspace) - SCIG Model (Simulink based)
```

```

t = 0:0.000002:1;
subplot(311);
plot(t,out.is_IG,'LineWidth',1.5);
xlabel("Time");ylabel("A");title("Stator Currents (ias,ibs,ics)");ylim([-30000 30000]);
subplot(312);
plot(t,out.Te_IG,'LineWidth',1.5);
xlabel("Time");ylabel("Nm");title("Electromagnetic Torque (Te)");ylim([-40000 40000]);
subplot(313);
plot(t,out.nm_IG,'LineWidth',1.5);
xlabel("Time");ylabel("RPM");title("Rotor Speed (nm)");ylim([1440 1520]);
stepinfo(out.nm_IG,t)

```

```
%Time for Array (To Workspace) - SCIG Model (Simscape Based)
```

```

t = 0:0.000002:1;
subplot(311);
plot(t,out.is_IG2,'LineWidth',1.5);

```

```
xlabel("Time");ylabel("A");title("Stator Currents ias,ibs,ics");ylim([-30000 30000]);  
subplot(312);  
plot(t,out.Te_IG2,'LineWidth',1.5);  
xlabel("Time");ylabel("Nm");title("Electromagnetic Torque (Te)");ylim([-40000 40000]);  
subplot(313);  
plot(t,out.nm_IG2,'LineWidth',1.5);  
xlabel("Time");ylabel("RPM");title("Rotor Speed (nm)");ylim([1440 1520]);  
stepinfo(out.nm_IG2,t)
```

In plot code, as per requirement values code will be commented and plot will be displayed.